

西南财经大学本科期末考试试题册（B）

（2023—2024学年第1学期）

以下各项由命题教师填写：

课程名称：面向对象程序设计（JAVA SE）(英)

命题教师：王磊

适用对象（年级专业）：2023 信息管理与信息系统专业

使用试题的任课教师姓名：王磊

试题说明：

- 1、考试类型：闭卷[] 开卷[☒]
- 2、本套试题共 2 道大题，共 7 页，完卷时间 120 分钟。
- 3、考试用品中除纸、笔、尺子外，可另带的用具有：
计算器[] 字典[] 等
(请在下划线上填上具体数字或内容，所选[]内打钩)

考试时间（由制卷方填写）：

以下各项由学生填写：

任课教师：

年级专业：

学生姓名：

学 号：

- 考生注意事项：
- 1.出示学生证（或身份证）和准考证于桌面左上角，以备查验。
 - 2.拿到试卷后清点并检查试卷页数，如有重页、页数不足、空白页及印刷模糊等举手向监考教师示意调换试卷。
 - 3.答题前先将试题册及答题纸上的任课教师、专业、年级、学号、姓名填写完整。
 - 4.所有答案均需填写在答题纸上，答在试题册上无效。
 - 5.考生不得携带任何通讯工具进入考场。
 - 6.考试结束后，将试题册、答题纸和草稿纸等下发材料上交监考教师，不得带离考场。
 - 7.严格遵守考场纪律。

Part I: Single Choice Questions:

Mark your answers on below sheet (total 30 pts, 3 pts each)

Answer sheet:

1	2	3	4	5	6	7	8	9	10

1. Which one is not the **primitive** data types of Java?

- A. long
- B. char
- C. double
- D. String

2. Which of the following Java statements set variable a to **true** if n is even integer, and to false if n is odd integer? Assume that $n \geq 0$. (Note: even integers are those integers which have a remainder of 0, when divided by 2.)

- I. `boolean a = (n%2 %2==0);`
- II. `boolean a = (n % 2 == 0);`
- III. `boolean a = (n / 2 == 0);`
- IV. `boolean a = (n % 2 == n/2);`

- A. I only
- B. I and II only
- C. II and III only
- D. III and IV only

Read the following Java code fragment, answer the next 2 questions.

```
int sum = 0;
int i = 0;
while (i < 5) {
    sum = sum + i;
```

```
        i++;  
    }  
    System.out.print(i);  
    System.out.print(" ");  
    System.out.print(sum);
```

3. What is the value of i when System.out.print(i) is executed?

- A. 6
- B. 5
- C. 4
- D. 3

4. What is the value of sum when System.out.print(sum) is executed?

- A. 6
- B. 10
- C. 15
- D. 21

5. What is the output from the following Java program fragment?

```
public static void main(String[] args)  
{  
    int A = 10;  
    int B = 20;  
    update(A, B);  
    System.out.println(A + " " + B);  
}  
public static void update (int X, int Y)  
{  
    X = X + Y;  
    Y = Y + X;
```

```
System.out.println(X + " " + Y);  
}
```

A. 10 20

10 20

B. 10 20

30 50

C. 30 50

10 20

D. 30 50

30 50

6. After executing the following Java code, what is output?

```
public class FinalExam  
{  
    public static void main(String[] args) {  
        // Declare and allocate array  
        double array[] = new double[5];  
        // Initialize array  
        for (int index = 0; index < array.length; index++)  
            array[index] = (index * 3.0) + 0.5;  
        // Print array information  
        System.out.printf("%d", array.length - 2);  
        System.out.printf("%6.2f", array[2]);  
        System.out.printf("%6.2f",array[3]* 2.0);  
    }  
}
```

A. 3 6.5 19.00

B. 3 3.5 13.00

C. 2 6.5 19.00

D. 2 3.5 13.00

7. If a class is not qualified as public or private, what does that imply about its public methods and fields?

- A. No other class can use them: class access defaults to private.
- B. Only classes in the same package can use them.
- C. Only classes derived from this class can use them.
- D. All classes can use them.

Polygon is a class that defines regular polygons (figures such as equilateral triangles, squares, and regular pentagons—polygons where all the sides have the same length).

Read the following classes, and answer the next 3 questions.

```
class Polygon{
    public double side;
    public double area() {
        return 0;
    }
}
class EquTriangle extends Polygon{
    public double area() {
        return side*side*Math.sqrt(3)/4;
    }
}
class Square extends Polygon{
    public double area() {
        return side*side;
    }
}
class Pentagon extends Polygon{
}
```

8. Which among the classes **Square**, **EquTriangle** and **Pentagon** inherit the area() method of Polygon?

- A. Square, EkiTriangle and Pentagon
- B. Square and EkiTriangle only
- C. Pentagon only
- D. none of them

9. Define P to be a Polygon object reference, S a Square object reference and T an **EquTriangle** object reference. Which of these assignment statements are legal?

- I. S = T III. T = P
- II. P = T IV. P = S

- A. II and IV only
- B. II and III only
- C. II, III and IV only
- D. I, II and IV only

10. Suppose **EquTriangle** has a public method info(). In this method, if you wish to call the area() method of superclass (i.e., **Polygon**). Which statement is **correct**?

- A. (polygon)area();
- B. area();
- C. super.area();
- D. this.area();

Part II: Write Programs.(total 70 pts)

1. (10pts) Write a program which generates 100 random integers between -10 and 10 and then displays the numbers of positive integers, negative integers and 0 respectively.

2. (15 pts) Suppose an integer array contains 10 integer numbers. Please complete the following **uniqueArray**() method to display unique numbers of it in order in the Console window. (e.g., array {1, 6, 10, 1, 1, 2, 3, 6, 4, 5}) has unique numbers 1,2,3,4,5,6,10).

```
public static void uniqueArray(int [ ] x){  
  
    }  
}
```

3. (20 pts) Design an abstract class **Solid** to describe all 3D geometries, which contains the below members:

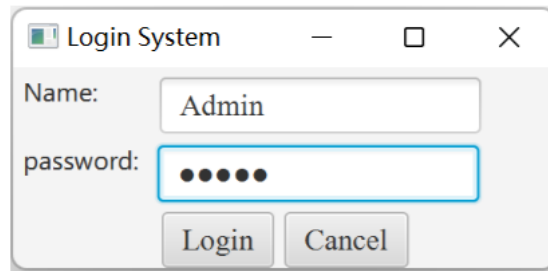
- ✓ An abstract method, **getVolume**(), returns the volume of the solid object.
- ✓ An abstract method, **compareTo**(Solid x), compares the volume of current solid object with the Solid object x. It will return an integer value (1 when current is larger than x, -1 when it smaller than x, and 0 when they have the same volume).

Then, you should design a **Cube** class to describe all 3D cuboid having the same side length, which extends from the **Solid** class. It contain the below members:

- ✓ A member variable named **side** denotes the length of any side of the Cube object.
- ✓ A constructor to initialize the member variables of the Cube object.
- ✓ Overridden method.
- ✓ A **toString**() method to display basic information of the Cube object, at least including its side length, volume.

Finally, you should create two **Square** objects with side length of 5 and 6 respectively, and compare their volumes, as well as display their basic information.

4. (25 pts) Please design a Java program to authorize users to login in an Information System, as shown in below figure. You should note that only one login name, Admin, is allowed to login in the System with password X1234.



Requirements:

- Use suitable Java FX GUI components.
- Use suitable layout to organize components.
- Contain at least one event handling process.